Chapter 1

Introduction

1.1 Background

The rapid growth of distributed energy resources (DERs) is increasing the need for more flexible distribution network operation. Traditional fixed export limits can be overly conservative and may restrict the efficient utilisation of DERs.

1.2 Research Motivation

Operating Envelopes (OEs) provide time-varying and customer-specific export and import limits based on network conditions. However, there remains limited understanding of what drives OE variability, which constraints most frequently limit OEs, and how much benefit OEs provide compared with fixed export limits.

1.3 Research Objectives

The objectives of this dissertation are:

1. Quantify export and import operating envelopes using a linearized optimal power flow framework.

2. Identify feeder and DER characteristics that most strongly influence the magnitude and variability of operating envelopes.

3. Determine the conditions under which customers experience large disparities in feasible DER operation.

4. Quantify the benefit of dynamic operating envelopes compared with traditional fixed export limits.

1.4 Research Contributions

The expected contributions of this work include:

- Development of a linearized DistFlow-based OE calculation framework;

- Large-scale OE computation using Hydro-Québec distribution feeder data;
- Characterisation of spatial and temporal OE behaviour;
- Identification of key OE drivers through statistical and machine-learning methods;
- Quantification of customer disparity and dynamic OE benefits.

Chapter 2

Literature Review

2.1 Operating Envelopes in Distribution Networks

- Motivation for operating envelopes
- Fixed export limits
- Dynamic operating envelopes
- Export and import operating envelopes
- Fairness and OE allocation

2.2 DER Flexibility Characterisation

Based on the literature on DER flexibility:

- Capability
- Feasibility
- Technical flexibility
- Commercial flexibility

Operating envelopes are interpreted as network-constrained feasible flexibility.

2.3 Optimal Power Flow Formulations for Distribution Networks

- AC Optimal Power Flow
- DistFlow formulation
- Linearized DistFlow
- Convex relaxations

2.4 Three-Phase Unbalanced Network Modelling

- Sources of network unbalance
- Single-phase DER integration
- Three-phase power flow models
- Impact of unbalance on OE calculation

2.5 Research Gap

Chapter 3

Methodology

3.1 Linearized Three-Phase Optimal Power Flow Model

- DistFlow equations
- Linearized DistFlow formulation
- Three-phase extension
- Modelling assumptions
- Solver implementation

3.2 Network Constraints Formulation

- Voltage constraints
- Thermal loading constraints
- Transformer constraints
- DER operating constraints
- Unbalance constraints

3.3 Operating Envelope Computation Framework

- Export OE formulation
- Import OE formulation
- Customer-specific optimisation
- Time-varying OE calculation

3.4 Forecast Input Assumptions

- Passive demand forecasts
- DER generation forecasts
- Forecast uncertainty considerations

3.5 Constraint Identification Framework

Identification of active constraints through:

- Voltage upper limits
- Voltage lower limits
- Thermal loading limits

3.6 Customer Disparity Metrics

- Range
- Standard deviation
- Maximum-to-minimum ratio
- Quantile-based measures

3.7 Data-Driven Characterisation Framework

Input features:

- Distance from substation
- Feeder depth
- Electrical distance
- Local demand
- Net demand
- PV generation
- DER capacity
- Network loading

Methods:

- Correlation analysis
- Regression analysis
- Random Forest feature importance

3.8 Fixed Export Limit Benchmark

Comparison metrics:

- Total exportable energy
- Curtailment reduction
- Network utilisation

Chapter 4

Case Study Network and Data

4.1 Hydro-Québec Distribution Feeder

4.2 Network Characteristics

- Feeder topology
- Line parameters
- Customer locations
- DER penetration

4.3 Demand and DER Forecast Data

- Passive customer demand forecasts
- DER generation forecasts

4.4 Data Processing and Preparation

4.5 Model Validation

Chapter 5

Results I: Impact of Feeder and DER Characteristics on Operating Envelopes

5.1 Baseline Operating Envelope Profiles

5.2 DER Capacity Sensitivity Analysis

5.3 DER Location Sensitivity Analysis

5.4 Time-Varying Operating Envelope Behaviour

5.5 Multi-DER Penetration Scenarios

Chapter 6

Results II: Constraint Binding and Customer Disparity

6.1 Binding Constraint Identification

6.2 Spatial Disparity Analysis

6.3 Temporal Disparity Analysis

6.4 Conditions Leading to Large Disparities

6.5 Feature Importance Analysis

Chapter 7

Results III: Benefits of Dynamic Operating Envelopes

7.1 Fixed Export Limit Benchmark

7.2 Total Exportable Energy Comparison

7.3 Curtailment Reduction Analysis

7.4 Per-Customer Benefit Distribution

7.5 Practical Implications

Chapter 8

Conclusions and Future Work

8.1 Main Findings

8.2 Key Drivers of Operating Envelopes

8.3 Customer Disparity Findings

8.4 Implications for Hydro-Québec

8.5 Limitations

8.6 Future Research Directions